

Philip Wang

Professor Adisa

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Understanding Crustacean Pain: Ethical Considerations in Aquatic Welfare

In recent years, the ethical treatment of animals has become an increasingly prominent topic. Discussions have extended beyond traditional practices regarding vertebrates, like dogs and cats. The conversation now includes invertebrates, even the seemingly lesser-known inhabitants of aquatic ecosystems, such as crustaceans. This diverse and unique group of organisms has been often overlooked within ethical circles. As our scientific understanding of these creatures continues to deepen, evidence suggests that crustaceans do, in fact, experience pain and distress. This discovery calls into question the conventional methods used to treat these creatures, particularly in the context of aquaculture and fisheries. If this is true, we cannot in good conscience hold on to our outdated ways of handling these aquatic organisms. New policies designed to promote ethical and compassionate treatment of crustaceans would need to be adopted.

According to the International Association for the Study of Pain, pain is defined as “an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage” (International Association for the Study of Pain). Both humans and crustaceans are able to detect noxious stimuli, such as heat, mechanical pressure, or chemicals through nociceptors, which are nerve endings that initiate reflexes that cause the swift withdrawal of the animal from the site of the stimulus (Elwood). In humans, however, exposure

to noxious stimuli generally leads to a negative state that we consider to be pain. A frequent argument against pain in crustaceans is that their general response to noxious stimuli is a mere nociceptive reflex (Elwood). If such reflexes are not the result of a crustacean's central nervous system, the common belief is that these reflexes would not register as pain (Elwood). The perception of pain is a critical element to our discussion. If crustaceans do not experience pain, then there would be very few welfare concerns regarding their suffering. It should be clarified that although the initial response to tissue damage in crustaceans might involve a reflex, a subsequent pain experience is still possible, which is also observable in humans. Assessing the experience of pain is challenging because immediate withdrawal reactions cannot be relied on in order to suggest physical pain. Pain is also a private experience, and crustaceans cannot report their feelings through communication, especially if pain is beyond just the physical. As a result, it is important to establish baseline criteria to determine the experience of pain in crustaceans.

According to several recent scientific studies, there are three main observable behaviors that can positively suggest the experience of pain. As summarized by Elwood, these include: (i) swift avoidance learning and aversion of anxiety; (ii) movement or activity directed specifically towards the site of damage and reduction in the use of certain appendages; and (iii) protection from further damage through the process of limb autotomy (Elwood). All of these behavioral responses go beyond simple nociceptive reflexes and form the necessary baseline conditions needed to assess whether crustaceans feel pain.

Crustaceans tend to demonstrate swift avoidance learning when repeatedly exposed to noxious stimuli. In one scientific study, researchers at Yamagata University used marbled crayfish to demonstrate this behavior. When placed in a T-maze (see Fig. 1 below), the marbled crayfish initially showed a preference for a chamber with blue light compared to a chamber with

white light (Okada). However, when these scientists applied an electric shock in the chamber with the blue light, the crayfish immediately switched their preference to the white-light chamber after just one electric-shock exposure in the blue-light chamber (Okada). Increasing the number of electric shock exposures resulted in longer avoidance periods. For example, after three exposures to electric shock, the crayfish refrained from entering the blue-light chamber well beyond 48 hours (Okada). The researchers concluded that the avoidance characteristics demonstrated by the crayfish was indicative of discomfort and anxiety associated with the noxious stimuli (e.g., electric shock) (Okada).

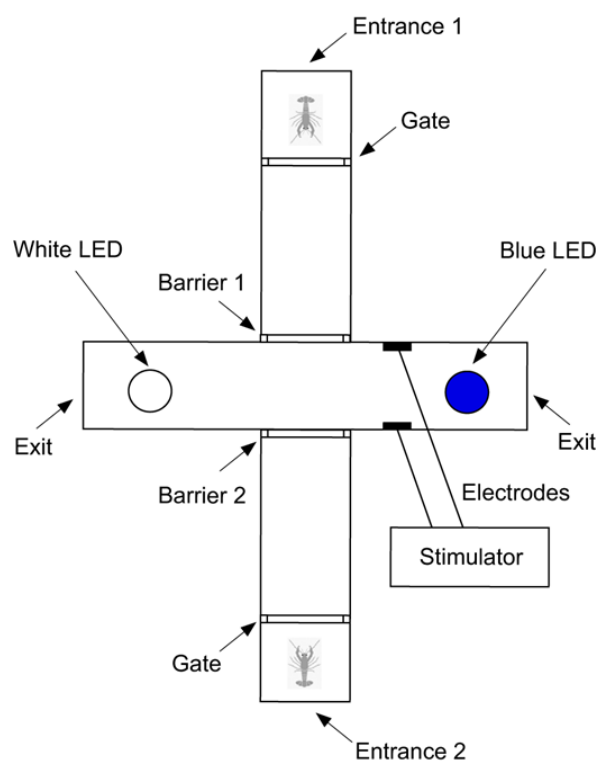


FIG. 1 - T-Maze Experiment Designed by Researchers at Yamagata University

In another experiment using hermit crabs that were given an electric shock, these hermit crabs would evacuate their shells temporarily (Appel and Elwood). Interestingly, when offered a new shell within 20 seconds, the shocked crabs were more likely than non-shocked crabs to

approach and move into the new shells (Appel and Elwood). Similar to the T-maze experiment, these hermit crabs demonstrated swift avoidance learning.

In addition to avoidance learning, the acts of rubbing an injury, guarding wounds, and limping are all behavioral actions that are consistent with the experience of pain (Wuertz). Scientists have observed these responses in crustaceans as well. In one scientific study, when sodium hydroxide and acetic acid (both known to induce pain in mammals) were applied to only a single antenna of a glass prawn, the prawn exhibited prolonged grooming behavior by repeatedly rubbing and pulling, using its small claws and mouthparts, that one antenna affected by the acid (Elwood). In fact, the glass prawn paid more attention to grooming the antenna treated with the acid than the untreated antenna (Elwood). This clearly indicates a high level of injury awareness, or at least the ability to locate the site of the noxious stimulus. Similar activity was observed in other crustaceans, such as crabs. In one study, rather than grooming, shore crabs used their claws to scratch at their mouthparts that had been brushed with acetic acid (Elwood). In another study, when crabs that had a clawed appendage twisted off (a common practice in some fisheries), these crabs positioned and held their remaining claw over the wound site (Elwood). In yet another experiment, hermit crabs that received an electric shock often directed their actions towards their wound site, such as evacuating their shells, grooming the body part where they were shocked, or simply moving away from the source of the shock (Appel and Elwood). Putting it all together, crustaceans demonstrate specific behaviors or activity indicative of a pain experience when they groom, rub, or perform related actions towards the site of injury or loss of body part or appendage.

Another potential indication of pain is through any behavioral response that seeks to mitigate pain. In addition to some of the aforementioned avoidance patterns, one such mitigation

technique employed by crustaceans is commonly referred to as autotomy. If a crustacean, for example, experiences trauma or injury in one of its appendages, it may autotomize (cut off) that limb to relieve itself of any potential and prolonged pain experience. The process of autotomy in crustaceans is quite elegant. Through the process of muscular contraction, crustaceans are able to disjoin their limb from a predetermined fracture plane. This is similar to some reptiles or amphibians that can autotomize a tail that has been ensnared by a predator. Although crustaceans are not known to regenerate lost body parts, any limb that is autotomized can quickly heal itself at the joint.

Autotomy can be reliably induced in crustaceans. In one experiment, the surface of the distal joint of a crab claw was wounded, causing a major loss of hemolymph (the crustacean equivalent to blood). Here, the crab was observed to reliably trigger a response to that injury by autotomizing the damaged claw (Elwood). Even in injuries that do not cause severe hemolymph loss, autotomy has also been observed after the claws of shore crabs were injected with formalin. Formalin is a solution that contains formaldehyde, which in high concentrations can be highly toxic, and in this case, functions as a noxious stimulus for the test subject crabs (Elwood). Furthermore, in other studies, it has been shown that crabs have also autotomized their limbs when the whole animal was placed on a hot plate or if the leg was subjected to electric shock (Elwood).

These studies and experiments all seem to suggest that crustaceans experience pain. Their observable behaviors clearly meet each and every one of the baseline criteria listed above, debunking the traditionally-held notion that their sudden movement to noxious stimuli is just mere reflex. Yet, despite the growing scientific knowledge that suggests that crustaceans can feel pain, very few solutions have been proposed to improve the ethical treatment of these organisms.

After numerous failed attempts, some organizations have been formed and are dedicated to the humane treatment of crustaceans and have tried to engage with legislators and policymakers in order to strengthen and enforce animal welfare law (Crustacean Compassion). Common practices in crustacean aquaculture, such as overcrowded conditions, transportation, and methods of harvesting, are indirectly contributing to heightened stress and pain in these organisms. High stocking densities, prevalent in many animal storage facilities, can lead to increased aggression, competition for resources, and the spread of diseases among crustaceans. Furthermore, the traditional practice of transporting crustaceans over long distances in sub-optimal conditions can induce pressure, compromising their overall welfare (Crustacean Compassion). Harvesting methods, including live storage and processing, may also subject crustaceans to avoidable stress and pain. As common as they are, these practices need to be reevaluated to meet ethical standards to align with the evolving understanding of crustacean sentience. Furthermore, public awareness of the ongoing crisis is extremely low, with many individuals largely unaware of the ethical considerations surrounding the treatment of these aquatic organisms (Crustacean Compassion). People's attitudes are often shaped by cultural norms, but the lack of attention to this issue involving crustaceans clearly contributes to a lack of empathy for their well-being.

However, it is never too late to act in response to this crisis. Take, for example, the French delicacy foie gras, which is the fattened liver of a duck or goose. The preparation process involves force-feeding the birds so that their livers expand to eight times the normal size. Because of increased attention to these deplorable practices, this dish was recently banned in over a dozen countries after finally being ruled inhumane (Animal Legal Defense Fund). In the same way, should scientific findings on crustacean pain perception become more widely

disseminated, there is greater possibility for increased public interest and concern regarding the ethical treatment of crustaceans in aquaculture. Innovations in aquaculture technology are being developed in order to reduce the impact of human interventions on crustacean welfare. For example, there have been recent research directed to low-stress handling and transportation of crustaceans and the like. Moreover, implementation of gradual acclimatization processes, improved water quality management, and the provision of enriched environments are being explored as strategies to enhance crustacean well-being. Additionally, the investigation of anesthesia and analgesia in crustaceans represents a promising avenue to mitigate pain (Crustacean Compassion).

The ongoing exploration of crustacean pain perception and its ethical implications reveals a nuanced and evolving field that demands our immediate attention and consideration. As with many issues, there are skeptics that continue to deny these recent findings. Yet, even non-experts can see that this is not the case. In a personal account about a trip to the Lobster Festival in Maine, journalist and writer David Foster Wallace states that, “you can usually hear the cover [of a pot] rattling and clanking as the lobster tries to push it off,” and that crustaceans behave “very much as you and I would behave if we were plunged into boiling water” (Wallace).

With several recent studies demonstrating how crustaceans feel pain in response to damaging stimuli, it is becoming evident that a more comprehensive understanding of crustacean welfare is essential. We must fulfill our ethical responsibilities by striving to integrate considerations of crustacean welfare into industry norms and regulations. This integration is a necessary step towards ensuring the well-being of crustaceans and other similar organisms, acknowledging their sentience, and ultimately fostering a more compassionate interaction between humanity and these distinct inhabitants of our ecosystems.

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